

ESAT CAMP MOCK TEST E

Physics

EC-PHY-E

PHYSICS

40 MINUTES

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Physics

- 1 A solid block of uniform cross-sectional area A and mass m floats upright in a liquid of density ρ_0 , as shown in the diagram. The lower portion of the block is submerged to a depth h .

The liquid is then heated uniformly. The liquid expands so that its density ρ_0 decreases, while the thermal expansion of the solid block is negligible. The block remains floating upright.

Here are three statements about the system after the liquid is heated:

- 1 The submerged depth h of the block increases.
- 2 The upthrust (buoyant force) acting on the block decreases.
- 3 The liquid pressure acting on the bottom surface of the block remains unchanged.

Which of these statements is/are correct?

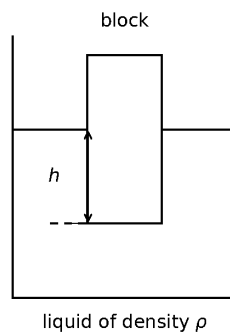


Diagram not to scale

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 2 A radioisotope thermal generator uses a radioactive isotope P, which decays into a single stable isotope Q. The heat generated is proportional to the activity of the sample.

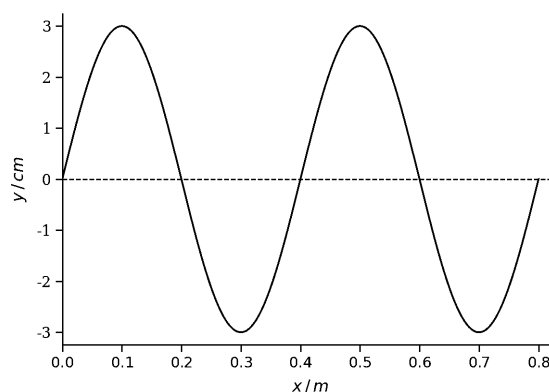
The fuel is initially pure isotope P, which has a half-life of 20 years. After 80 years of operation, what is the ratio of the number of atoms of Q to the number of atoms of P?

- A 1/16
- B 1/15
- C 7
- D 15/16
- E 15
- F 16

- 3 A continuous transverse wave travels at a constant speed of 12 m s^{-1} along a stretched string in the positive x -direction. The graph shows the displacement y of the string as a function of position x at time $t = 0$.

What is the total distance travelled by a particle of the string during a time interval of 0.50 s, and what is the distance travelled by the wave during this same time interval?

	distance travelled by particle / m	distance travelled by wave / m
A	0.45	6.0
B	0.45	12.0
C	0.90	6.0
D	0.90	12.0
E	1.8	6.0
F	1.8	12.0
G	3.6	6.0
H	3.6	12.0



- 4 An electric heater supplies energy at a constant rate of 400 W to a metal block of mass 3.0 kg. The specific heat capacity of the metal is $1000 \text{ J kg}^{-1} \text{ K}^{-1}$. The block loses heat to its surroundings at a constant rate of 100 W.

What is the time taken for the temperature of the block to increase by 20 K?

- A 120 s
- B 150 s
- C 200 s
- D 300 s
- E 600 s

- 5 Spherical catalyst particles of radius r fall through a gaseous reactant mixture. The drag force F acting on a particle moving with velocity v is given by:

$$F = kr^2v^2$$

where k is a constant.

The mass of a particle is proportional to r^3 . At the terminal velocity v_t , the drag force is equal to the gravitational force.

Which of the following gives the relationship between v_t and r ?

- A $v_t \propto r^2$
- B $v_t \propto r$
- C $v_t \propto \sqrt{r}$
- D $v_t \propto \frac{1}{\sqrt{r}}$
- E $v_t \propto r^{\frac{3}{2}}$

- 6 An astronaut of mass 100 kg is at rest in space. The astronaut throws a 4.0 kg tool, which then moves at a speed of 5.0 m/s. What is the astronaut's recoil speed?

- A 0.05 m/s
- B 0.20 m/s
- C 1.0 m/s
- D 5.0 m/s
- E 20 m/s
- F 125 m/s

- 7 A U-shaped magnet rests on an electronic balance. A rigid, horizontal straight wire of length 15 cm is clamped independently in a fixed position between the poles of the magnet, perpendicular to a uniform magnetic field of flux density 0.40 T.

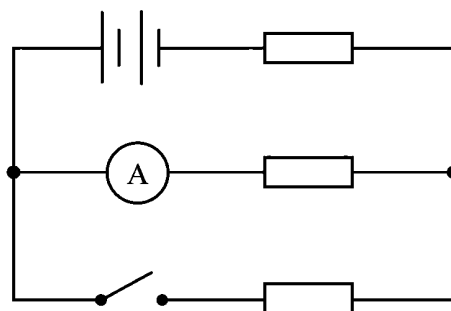
A constant current is switched on in the wire such that a total charge of 45 C passes through the wire in 2.5 minutes. This current causes a vertically upward magnetic force to act on the wire.

What is the resulting change in the reading displayed by the balance?

(Take the acceleration of free fall to be $g = 10 \text{ m s}^{-2}$.)

- A decrease of 0.18 g
- B increase of 0.18 g
- C decrease of 1.8 g
- D increase of 1.8 g
- E decrease of 18 g
- F increase of 18 g
- G decrease of 108 g
- H increase of 108 g

- 8 A circuit is set up as shown. The battery has negligible internal resistance and all three resistors are identical.



When the switch is open, the reading on the ammeter is 3.0 A and the power transferred from the battery is 18 W.

The switch is now closed.

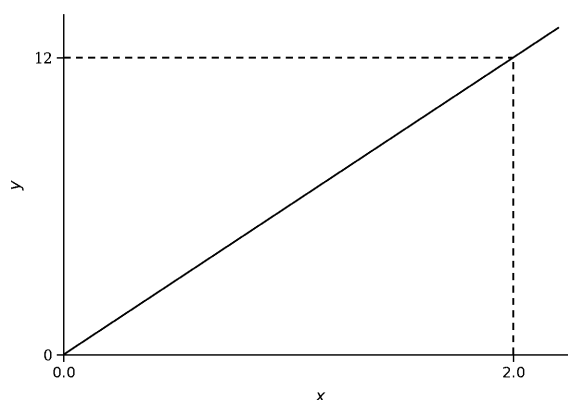
What is the new reading on the ammeter and what is the new power transferred from the battery?

	ammeter reading / A	power transferred / W
A	1.5	18
B	1.5	24
C	2.0	12
D	2.0	24
E	2.0	36
F	3.0	24
G	3.0	27
H	3.0	36

- 9** The graph shown of quantity y against quantity x represents a physical relationship, where both quantities are measured in their standard SI units.

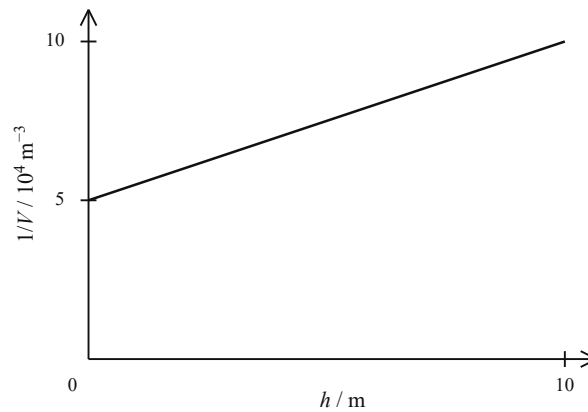
Which two of the following relationships could this graph represent?

- 1 momentum against velocity for an object of mass 6.0 kg
- 2 kinetic energy against velocity for an object of mass 12 kg
- 3 electrical energy transferred against charge transferred across a component with a constant potential difference of 6.0 V
- 4 tension against extension for a system of two identical springs connected in series, each having a spring constant of 6.0 N m^{-1}



- A** 1 and 2
 - B** 1 and 3
 - C** 1 and 4
 - D** 2 and 3
 - E** 2 and 4
 - F** 3 and 4
- 10** A drone travels in a straight line between two points separated by a total distance of 800 m. For the first distance d , it travels at a constant speed of 10 m s^{-1} . For the remainder of the journey, it travels at a constant speed of 25 m s^{-1} . The average speed for the entire journey is 16 m s^{-1} . What is the value of d ?
- A** 200 m
 - B** 250 m
 - C** 300 m
 - D** 400 m
 - E** 500 m
 - F** 600 m

- 11 An air bubble rises through water of constant temperature. A graph of the reciprocal of the bubble's volume, $1/V$, against its depth, h , is a straight line as shown.



What is the atmospheric pressure at the surface of the water?

(Use: density of water, $\rho = 1000 \text{ kg m}^{-3}$; gravitational field strength, $g = 10 \text{ N kg}^{-1}$)

- A $1.0 \times 10^5 \text{ Pa}$
- B $2.0 \times 10^4 \text{ Pa}$
- C $5.0 \times 10^4 \text{ Pa}$
- D $2.0 \times 10^5 \text{ Pa}$
- E $1.0 \times 10^4 \text{ Pa}$
- F $5.0 \times 10^5 \text{ Pa}$

- 12** An LDR and a $3.0\text{ k}\Omega$ resistor are connected in series across a 12 V supply, as shown. A control unit switches on when the voltage across the fixed resistor exceeds 8.0 V . The control unit draws negligible current.

At what LDR resistance is the voltage exactly 8.0 V , and which change will then switch the control unit on?

- A** $1.5\text{ k}\Omega$; increasing light intensity
 - B** $1.5\text{ k}\Omega$; decreasing light intensity
 - C** $2.0\text{ k}\Omega$; increasing light intensity
 - D** $6.0\text{ k}\Omega$; increasing light intensity
 - E** $6.0\text{ k}\Omega$; decreasing light intensity
- 13** A student performs a calorimetry experiment using a shiny copper beaker containing hot water. To improve the accuracy of the enthalpy change determination, the student considers several modifications to reduce heat loss to the surroundings.
- Which of the following modifications would be **counter-productive** (i.e., would increase the rate of heat loss)?
- A** Coating the outer surface of the beaker with a thin layer of dull black paint
 - B** Polishing the outer surface of the beaker to a mirror finish
 - C** Placing a plastic lid on top of the beaker
 - D** Placing the beaker on a cork mat
 - E** Wrapping the sides of the beaker in mineral wool

- 14** A spring has an original length of 10.0 cm. It is stretched beyond its elastic limit. The graph shows the force as the spring is then unloaded.

What is the spring's length when the force has fallen to 2.0 N, and what is its permanent increase in length after complete unloading?

- A** 11.0 cm; 2.0 cm
- B** 13.0 cm; zero
- C** 13.0 cm; 2.0 cm
- D** 14.0 cm; 2.0 cm
- E** 16.0 cm; 6.0 cm

- 15** A pure substance is heated at constant power. The graph shows its temperature. During the horizontal section, solid and liquid are both present. Energy losses are negligible.

Which statement is correct during this horizontal section?

- A** The average kinetic energy of its particles stays constant while their potential energy increases.
- B** The average kinetic energy of its particles increases while their potential energy stays constant.
- C** No energy is transferred to the substance.
- D** Both the average kinetic energy and potential energy of its particles decrease.
- E** The particles stop moving until melting is complete.

- 16** Two identical, inverted test tubes, X and Y, contain trapped air. Tube X has a total mass m_X and tube Y has a total mass m_Y , where $m_X > m_Y$.

Both tubes are submerged in water and come to rest, neutrally buoyant, at depths h_X and h_Y respectively.

The temperature is constant throughout.

Which statement correctly compares the depths h_X and h_Y ?

- A** $h_X < h_Y$
 - B** $h_X > h_Y$
 - C** $h_X = h_Y$
 - D** The relationship depends on the specific ratio m_X/m_Y .
 - E** The relationship depends on the value of atmospheric pressure.
 - F** There is not enough information to determine the relationship.
- 17** A resistor of constant resistance dissipates 36 W when the potential difference across it is 12 V. The potential difference is changed to 8 V. What is the power dissipated by the resistor at this new potential difference?
- A** 16 W
 - B** 24 W
 - C** 36 W
 - D** 54 W
 - E** 81 W

- 18** A straight horizontal metal rod lies in the plane of the page in a uniform magnetic field directed into the page. The rod is moved without rotating.

Consider these motions:

1. vertically upwards in the plane of the page;
2. horizontally to the right, along the rod's own length;
3. directly into the page, parallel to the magnetic field.

For which motions is a voltage induced between the two ends of the rod?

- A** 1 only
- B** 2 only
- C** 3 only
- D** 1 and 2 only
- E** all three

- 19** An object of constant mass m is in motion. A graph of its kinetic energy against the square of its momentum (p^2) is plotted.

The graph is a straight line passing through the origin with gradient G .

What is the mass m of the object in terms of G ?

- A** $\frac{1}{2G}$
- B** $\frac{1}{G}$
- C** $\frac{G}{2}$
- D** $2G$
- E** $\frac{2}{G}$
- F** G

- 20** A source and detector monitor the thickness of a continuous sheet of paper. The source must work for several years without a large correction for radioactive decay. Alpha radiation is completely stopped by the paper; beta radiation is partly absorbed; gamma radiation is hardly affected by changes in the paper's thickness.

Which source is most suitable, and what happens to the detector count rate if the sheet becomes thinner?

- A** alpha, half-life 20 years; decreases
 - B** beta, half-life 2 hours; increases
 - C** beta, half-life 20 years; increases
 - D** beta, half-life 20 years; decreases
 - E** gamma, half-life 20 years; increases strongly
- 21** A wave travels from medium P to medium Q. In medium P, the wave speed is 400 m/s and the wavelength is 0.60 m. In medium Q, the wave speed is 1200 m/s.

What is the wavelength of the wave in medium Q?

- A** 0.20 m
 - B** 0.60 m
 - C** 1.80 m
 - D** 5.40 m
- 22** A particle is projected vertically upwards on a planet. In the absence of an atmosphere, the particle would reach a maximum height H_0 .

In the presence of an atmosphere, the particle experiences a constant resistive force of magnitude $\frac{1}{4}$ of its weight.

Which expression gives the maximum height H reached by the particle in the presence of the atmosphere?

- A** $\frac{3}{4}H_0$
- B** $\frac{4}{5}H_0$
- C** H_0
- D** $\frac{5}{4}H_0$
- E** $\frac{4}{3}H_0$

- 23** Paint droplets are given a negative charge before being sprayed towards a positively charged metal panel. The droplets stick to the panel and transfer their excess electrons to it.
- The panel is kept at a constant positive charge by an electrical connection. Which row describes the electric force between two droplets and the electron transfer needed through the connection?
- A** droplets attract; electrons enter the panel
 - B** droplets attract; electrons leave the panel
 - C** droplets repel; electrons enter the panel
 - D** no force between droplets; no electron transfer
 - E** droplets repel; electrons leave the panel
- 24** A magnet is moved along the axis of a stationary coil. Moving its north pole towards the coil gives a positive voltage at the voltmeter.
- In a second experiment, its south pole is moved towards the coil, held stationary for a short time, and then moved away. The voltmeter connections are unchanged.
- Which sequence describes the signs of the voltages during these three stages?
- A** negative, zero, positive
 - B** positive, zero, negative
 - C** negative, negative, positive
 - D** positive, zero, positive
 - E** negative, zero, negative

- 25** A light ray crosses a boundary from material X into material Y. The angles marked on the diagram are measured between the ray and the boundary.

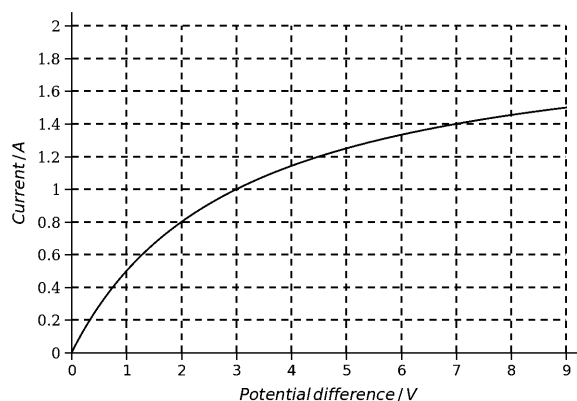
Which statement correctly compares the wave in Y with the wave in X?

- A** lower speed, lower frequency
- B** lower speed, same frequency
- C** higher speed, same frequency
- D** higher speed, higher frequency
- E** same speed, same frequency

- 26** A non-linear electrical component X has the current–potential difference ($I - V$) characteristic shown in the graph below.

Two identical components of type X are connected in parallel with each other. This parallel combination is connected in series with a $3.0\ \Omega$ resistor and an ideal d.c. power supply of electromotive force (EMF) $9.0\ \text{V}$ with negligible internal resistance.

What is the total power dissipated in the entire circuit?



- A** 3.0 W
B 6.0 W
C 9.0 W
D 12 W
E 15 W
F 18 W
G 24 W
H 27 W
- 27** A stationary sound source emits a continuous note of frequency $3300\ \text{Hz}$. A stationary observer is positioned some distance away from the source.
- A wind blows directly from the source to the observer at a constant speed of $30\ \text{m/s}$. The speed of sound in still air is $330\ \text{m/s}$.
- What is the frequency of the sound detected by the observer?
- A** 3000 Hz
B 3025 Hz
C 3300 Hz
D 3600 Hz
E 3630 Hz